Murphy: Data-Driven Flight Disruption Forecasts You Can Trust

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https://github.com/halleepham/murphy

GitHub

"Anything that can go wrong will go wrong." – Murphy's Law

"Forecasts contain no intrinsic value. They acquire value through their ability to influence the decisions made by users of the forecasts." – Allan H. Murphy

INTRODUCTION

The Problem:
- A flight delay is an inconvenience; a missed connection, wedding, or meeting is a real cost
- Travelers find out late, after the best rebooking options are gone

The Gap:
- Current tools give a delay prediction, but not how much to trust it or the evidence behind it

PROJECT GOAL

Use **machine learning**, **network analysis** and **evidence retrieval** on large-scale flight and weather data to produce **trustworthy disruption forecasts** and **recommendations** that help travelers decide whether to keep or change a flight.

DATA

26.3M	1.5M	1.44M
US flights, Jan 2023 – Aug 2026	weather reports and forecasts	forecasts scored for 2026

- **Flights:** BTS On-Time Performance, 7 airlines + regional partners
- **Weather:** IEM observations and forecasts as issued, 30 busiest airports
- **Fares:** BTS DB1B 2025 Q2, what people paid
- **Rules:** federal rules (14 CFR) and Delta, Southwest contracts

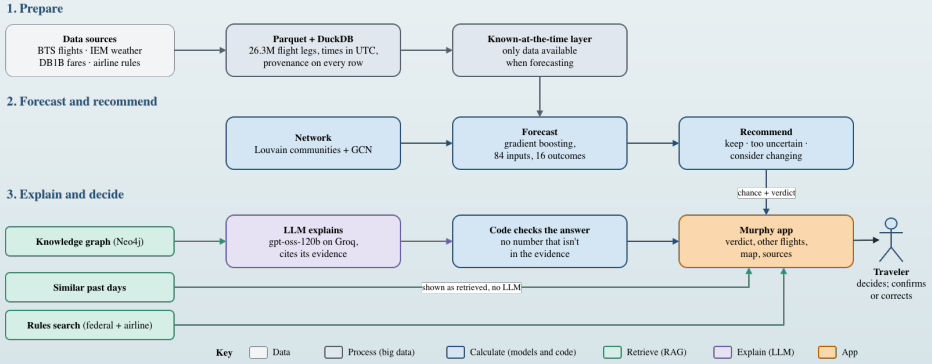
APP FEATURES

- **Verdict:** keep, too uncertain, or consider changing, with past accuracy
- **Deadline what-if:** drag the deadline and watch the chance change
- **Other flights:** chance, gain over yours, estimated ticket price; sort by price
- **Map:** delay risk by region, linked airports, weather, time slider
- **Similar days:** past days with this weather, and how they went
- **Rules:** what federal rules and the airline's contract say
- **Ask Murphy:** answers cite their evidence; confirm or correct them
- **Sources:** every answer shows the record, time and model it came from

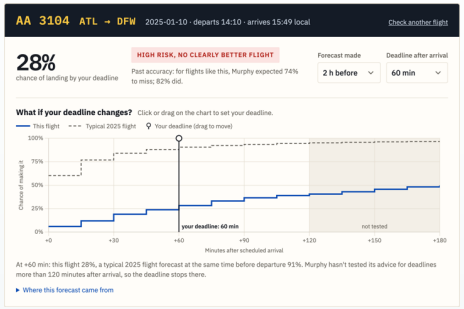
EVALUATION METHOD

- **Split:** train 2023–24, tune 2025, test Jan–Aug 2026 (sealed until the end and scored once)
- **No hindsight:** inputs only from before the forecast; leak checks
- **Forecast:** Brier score and calibration
- **Advice:** replay 1.44M forecasts on 480K sampled 2026 flights (24, 6 and 2 h before departure); count rescued vs. needless changes
- **Baselines:** every method vs. a simpler one, 95% intervals by day
- **Code:** automated tests, including failure cases

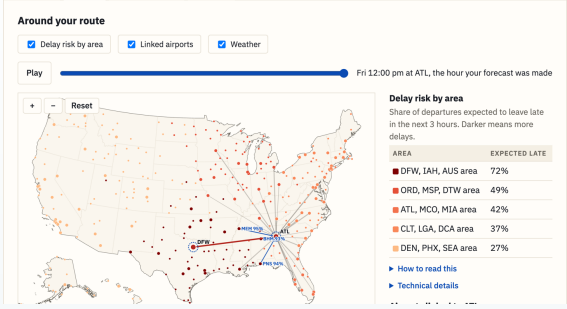
PROJECT PIPELINE



THE APP



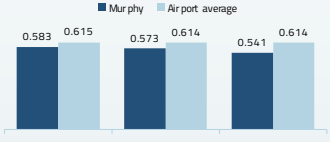
AA 3104 ATL → DFW, 10 Jan 2025 ice storm, forecast 2 h before departure



Same trip: delay risk by area, airports linked to ATL, weather, time slider

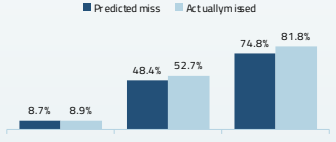
RESULTS

Forecast error, Brier (lower is better)



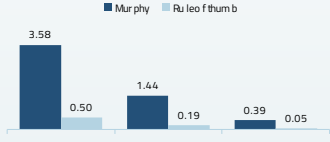
Better at every lead time

Track record, 60-min deadline



Close on keep; runs low on risky flights

Trips saved per 1,000, by deadline



Beats the rule of thumb at every deadline

3.1 : 1	**Rescued vs. needless changes** 60-min deadline, 2026
1.44M	**Forecasts scored once** Jan–Aug 2026, never seen in training
–0.049	**Forecast error vs. airport average** Brier gap; 2025 was –0.042

Every method vs. a simpler baseline

Task	Murphy	Simpler
Forecast error (Brier), 2026	**0.566**	airport average 0.614
Forecast error vs. schedule only, 2026	**0.566**	schedule only 0.602
Trips saved per 1,000 (60 min), 2026	**1.44**	rule of thumb 0.19
Similar days, error in late share	**8.2 pts**	month average 9.1
Rules: right passage in top 3	**14 of 16**	BM25 8
Answers with the key fact	**6 of 7**	BM25 5, none 0
Finding airports that go bad next	**45%**	today's order 30%

KEY FINDINGS

- **Holds up on an unseen year**: beats the airport average on 2026 by as much as on 2025
- **Advice is rare and earned**: keeps 96% of trips; 3 trips rescued per needless change
- **Honest uncertainty**: from 40% to 60% risk it says "too uncertain"; those flights missed 53% of the time
- **Hindsight kept out**: graph evidence is filtered by time; BM25 evidence leaked outcomes in 12 of 16 sets
- **Matching beats averages**: similar past days beat month averages
- **GCN**: finds airports going bad next (45% vs. 30%)

LIMITATIONS AND NEXT STEPS

- **Replay, not live**: Jan–Aug 2026 history; no fall or holidays
- **Runs low on risky flights**: 82% missed vs. 75% predicted
- **Fares are estimates**: median paid; no live fares
- **Small LLM test**: 4 flights; Groq's daily limit
- **Coverage**: weather at the 30 busiest airports only; direct flights only
- **No real users**: effect on decisions untested
- **Next**: fall and holidays; connections; user study

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